Supplement 1: Alternatives and supplements to ComBat for integrating microarray studies with non-overlapping biological groups

Motivation

ComBat [1] is the default empirical-Bayes method for removing batch effects from merged microarray data and is the engine used in the main pipeline. Its location-and-scale model assumes that each batch contributes additive and multiplicative perturbations that are *independent* of the biological covariates of interest. When the covariate of interest is balanced across batches, this assumption is harmless: the per-batch mean of the biological signal is the same, so subtracting per-batch means does not erase between-group contrasts.

The placental dataset collection used in this work violates that assumption in a hard way. Several Affymetrix and Illumina studies contain only a single trimester (e.g. first trimester only, or term only); other studies span two trimesters but never all three. There is no batch in which all three trimesters are observed jointly. Under this design, "batch" is partially aliased with "trimester": the batch-mean estimate that ComBat removes contains real biological signal, and the model has no way to separate the two without additional information.

Nygaard, Rødland, and Hovig formalised the problem and showed that running ComBat (or its parametric two-way ANOVA equivalent) on unbalanced designs and then handing the corrected matrix to a naive downstream test exaggerates between-group differences and inflates false positives, sometimes by an order of magnitude [2]. Their concrete recommendation is to skip the pre-correction step and instead include batch as a covariate in the differential-expression model directly, so that the residual degrees of freedom are charged correctly.

Towfic, Kusko, and Zeskind replied with a re-analysis of one of the datasets used by Nygaard (GSE61901), arguing that the inflation Nygaard reported was driven not by ComBat itself but by how *technical replicates* were handled in the analysis pipeline. When technical replicates were retained and modelled with limma's `duplicateCorrelation` or with mixed-effects regression rather than averaged away, several batch-handling strategies (including ComBat followed by limma) converged on a similar ~ 1500-DEG result, while the averaging strategy collapsed it to 11 DEGs [3]. The two papers therefore agree on the core diagnostic — when the design is unbalanced the choice of batch-handling pipeline materially changes the differential expression list and must be documented — but they disagree about which step in the pipeline is to blame: Nygaard fingers the ComBat-then-test sequence, Towfic et al. fingers the averaging of technical replicates.

This supplement reviews the methods that are most commonly used as *alternatives* or *supplements* to ComBat for this regime, groups them by the assumption they substitute for the "balanced design" assumption, and gives practical recommendations for the placenta integration use case. Downloaded references are listed in `references/integration_methods_references/`.

1 Why ComBat is fragile on unbalanced trimester data

ComBat fits, for every gene g and every sample i in batch b:

$$Y_{gi} \;=\; \alpha_g \;+\; X_i\,\beta_g \;+\; \gamma_{gb} \;+\; \delta_{gb}\,\varepsilon_{gi}, \tag{1}$$

where X_i encodes the biological covariates that should be preserved, γ_{gb} is the additive batch shift, and δ_{gb} is the multiplicative batch scale. Empirical-Bayes shrinkage of γ_{gb} and δ_{gb} across genes makes the fit stable even with few samples per batch.

The crucial step is that ComBat then *subtracts* the estimated $\hat{\gamma}_{gb}$ and divides by $\hat{\delta}_{gb}$ to produce a "corrected" Y^*_{gi} that downstream methods treat as if it came from a single batch. The estimate $\hat{\gamma}_{gb}$ is the within-batch residual mean after removing $X_i\,\beta_g$. If batch b contains only first-trimester samples, then "the within-batch mean after accounting for biology" is meaningless: the design matrix cannot identify β_g from γ_{gb} inside that batch alone, and the cross-batch pooling of $\hat{\gamma}_{gb}$ ends up importing trimester structure into the batch shift. The result is that the trimester effect is partially absorbed into the batch correction and then subtracted off.

Two observations follow:

1. If the goal is "one merged matrix that PCA, clustering, and visualisation can use," the bias from unbalanced ComBat is irreducible without strong external assumptions.

2. If the goal is differential expression, it is almost always safer to keep batch in the design matrix of the downstream linear model and *not* pre-subtract a batch shift [2, 3].

The methods reviewed below trade the "balanced design" assumption for one of: (a) the existence of negative-control genes, (b) the existence of a reference batch or replicate samples, (c) the existence of mutual-nearest-neighbour structure between batches, or (d) the willingness to combine effect sizes per study rather than samples.

2 Overview of direct-merge integration methods

The following table summarises the principal methods for direct-merge integrative analysis of gene-expression datasets, drawing on Tseng et al. [19], Yu et al. [20], Adamer et al. [21], and Baião et al. [22]. Methods are grouped by the taxonomy of Yu et al. into four categories: location-scale (LS), matrix-factorisation (MF), distance-neighbourhood (DN), and cross-platform harmonisation (CP). For each method we note whether it can handle confounded designs (batch aliased with biological group), which is the central problem in this work.

Method	**Category**	**Introduced**	**Latest biol. study**	**Handles confounded design?**	**Principle**	**Implementation**
ComBat	LS	2007	2025	No	Empirical Bayes location-scale; requires balanced groups across batches	`sva::ComBat`
ComBat + batch-in-design	LS	2016	2025	Partially	Standard ComBat followed by including batch as a covariate in the limma model; works when overlap is partial but not zero	`sva` + `limma`
Ref-batch ComBat	LS	2008	2024	Partially	ComBat with one batch held as reference; reference must contain all groups, others may be incomplete	`sva::ComBat (ref.batch=)`

Method	Category	Introduced	Latest biol. study	Handles confounded design?	Principle	Implementation
ComBat-seq	LS	2020	2025	No	Negative-binomial ComBat for RNA-seq counts; same confounding issue	`sva::ComBat_seq`
ComBat-ref	LS	2025	2025	Partially	Auto-selects reference batch by minimum dispersion; NB quantile matching; RNA-seq only	GitHub [10]
reComBat	LS	2022	2022	Partially	Regularised ComBat (elastic-net regression) for singular design matrices; handles highly correlated batch–biology designs	`reComBat` (Python) [21]
M-ComBat	LS	2015	2020	Partially	Modified ComBat that transforms all batches to a pre-determined reference batch	Custom R
Ratio-based scaling	LS	—	—	Yes	Scale features relative to concurrently profiled reference material; effective even when batch and biology are confounded [20]	Custom
`limma::removeBatchEffect()`	LS	2009	2025	No	Location-scale removal; same confounding limitation as ComBat	`limma`
HarmonizR	LS	2022	2025	Partially	Matrix dissection splits incomplete matrix into complete sub-matrices; applies ComBat or `removeBatchEffect()` to each independently; avoids imputation but inherits confounding limitations of the underlying method	`HarmonizR` (Bioconductor) [16]
SVA	MF	2007	2025	Partially	Estimates latent surrogate variables from residuals; risk of absorbing biology when fully confounded	`sva::sva`
RUV-2 / RUV-4 / RUVs	MF	2012	2025	Yes	Factor analysis on negative-control genes or replicate samples; does not need balanced groups	`ruv`, `RUVSeq`
RUV-III-PRPS	MF	2023	2023	Yes	Extends RUV-III by constructing pseudo-replicates when real controls are unavailable	`ruv`
MNN / fastMNN	DN	2018	2025	Yes	Aligns batches via mutual nearest neighbours; needs *some* biological overlap but not full factorial	`batchelor::fast`

Method	Category	Introduced	Latest biol. study	Handles confounded design?	Principle	Implementatio
Harmony	DN	2019	2025	Yes	Iterative soft-clustering alignment; originally single-cell, applicable to bulk	`harmony` (R/Python)
XPN	CP	2008	2024	No	Piecewise-linear cross-platform normalisation; only two datasets at a time; requires matched groups	`CONOR` (archived)
DWD	CP	2004	2024	No	SVM-like projection to remove platform shift; does not model biology	`dwd` (R)
CuBlock	CP	2021	2025	No	Iterative cubic fitting across matched clusters; requires shared structure	`CuBlock` (R)
Shambhala-1/2	CP	2019	2025	Yes	Uniformly-shaped harmonisation; each profile converted independently to a reference distribution shape	`Shambhala` (R)
fRMA	CP	2010	2023	Yes	Frozen RMA; pre-computed parameters from training data; each array processed independently	`frma` (Bioconductor)
TDM	CP	2016	2025	Yes	Quantile regression trained on a reference platform; transforms each target profile independently	`TDM` (R)
UPC	CP	2013	2013	Yes	Transforms each profile to a probability scale independently	`SCAN.UPC` (Bioconductor)
QN	CP	2003	2025	No	Quantile normalisation; equalises distributions but does not model batch vs. biology	`preprocessCore`
GQ / NorDi / DisTran / MRS	CP	2012	2012	No	Rank- or discretisation-based transformations; do not model batch–biology confounding	Various / custom
scGen	DL	2019	2022	Yes	Variational autoencoder; learns batch-free latent space; requires large sample sizes	`scgen` (Python)
normAE / AD-AE	DL	2020	2020	Yes	Adversarial autoencoder for batch removal; latent representation only	Python [21]

The key finding across reviews is that **no single method dominates** [19, 20]. For the placenta collection, where batch is partially or fully aliased with trimester, only methods marked

"Yes" in the confounding column are directly applicable without dropping single-trimester studies. Among these, RUV (requiring negative-control genes), ratio-based scaling (requiring reference material), Shambhala/fRMA/TDM/UPC (pre-defined-shape harmonisation), and Harmony/MNN (neighbourhood-based alignment) are the most practical choices.

2.1 Evaluation metrics for batch-effect correction

Assessing whether a batch-correction method has succeeded requires quantitative metrics beyond visual inspection. The following table summarises the evaluation approaches described in the overview literature [20, 21].

Metric	Category	Description
PCA / t-SNE / UMAP	Visualisation	Dimensionality-reduction plots coloured by batch and by biology; after correction, samples should cluster by biology, not batch [20]
Hierarchical clustering (HCA)	Visualisation	Dendrogram of samples; biology-driven clustering indicates successful correction [20]
Relative log expression (RLE)	Visualisation	Distribution of log-ratios to geometric mean per gene; batch-specific medians/variances indicate residual batch effects [20]
PVCA (Principal Variance Component Analysis)	Visualisation	Bar plot of variance attributable to batch, biology, and residual; combines PCA with variance-component analysis [20]
kBET (k-nearest-neighbour Batch Effect Test)	Distance	Chi-squared test on whether batch composition in local neighbourhoods matches global proportions; low rejection rate = well mixed [20]
LISI (Local Inverse Simpson's Index)	Distance	Diversity index in local neighbourhoods; high batch-LISI (batches mixed) + low biology-LISI (biology separated) = good correction [12, 20]
DRS (Distance Ratio Score)	Distance	Ratio of distance to nearest same-biology/different-batch sample vs. nearest different-biology sample; microarray-specific [20]
Shannon entropy	Distance	Entropy of batch labels in k-NN graph neighbourhoods; higher = better mixing [20, 21]
Alignment score	Distance	Examines local neighbourhood composition after alignment; scRNA-seq-focused [20]
Signal-to-noise ratio (SNR)	Distance	Ratio of between-group distance to within-group (technical replicate) distance in PCA space [20]
Guided PCA (gPCA)	Distance	SVD guided by batch indicator matrix to quantify and visualise batch effects [20]
ARI (Adjusted Rand Index)	Clustering	Agreement between true labels and clustering assignments; ARI by biology should be high, ARI by batch should be low [20]
ASW (Average Silhouette Width)	Clustering	Silhouette width by biology (high = good separation) and by batch (low = good mixing) [20]
Minimum separation number	Clustering	Minimum number of cluster merges needed to separate batch from biology; used by reComBat [21]
Cluster purity / Gini impurity	Clustering	Fraction of dominant label in each cluster; evaluated separately for batch and biology labels [21]
LDA score	Classifier	Linear discriminant analysis score for batch and biology labels; classifier-based metrics are the most informative according to Adamer et al. [21]

Metric	Category	Description
Balanced accuracy / F1 score	Classifier	Logistic regression or SVM trained to predict batch labels; low accuracy = batch signal removed [21]
Cross-dataset classification	Classifier	Train on Study A, predict biology labels in Study B; high accuracy = batch effect successfully removed [20]
R^2 (coefficient of determination)	Classifier	Proportion of variance in the dependent variable predictable from the independent variable; used to assess cross-dataset transferability [20]
DEG recovery (TPR, FPR, MCC)	Downstream	Compare detected DEGs against a ground-truth set (spike-in or known markers); true positive rate, false positive rate, Matthews correlation coefficient [20]
Fold-change correlation / RMSE	Downstream	Correlation or root-mean-square error of fold-changes vs. ground truth [20]
Cross-batch predictive modelling	Downstream	ROC/AUC, MSE, MAE, R^2 for a model trained on one batch and evaluated on another [20]
Correlation before/after	Downstream	Pearson/Spearman correlation of matched expression profiles before and after harmonisation [20]
Between/within class distances	Downstream	Comparison of distances between and within biological classes before and after correction [20]

2.2 Validation methods for confounded (batch-aliased-with-biology) designs

When batch is partially or fully aliased with the biological factor of interest — as in the placenta collection where some studies contain only one trimester — many standard evaluation metrics become misleading (e.g. kBET fails on confounded batch proportions [20]; batch silhouette width alone penalises legitimate biological separation; PCA visual inspection is qualitative and insufficient for reviewers).

The following methods are specifically suited for this scenario:

1. **PVCA (Principal Variance Component Analysis).** Decomposes total variance into batch, biology, and residual components via PCA + mixed-model variance partitioning. Show bar plots before and after correction: batch variance should decrease while trimester variance is preserved. *Popularity:* high; standard in microarray/RNA-seq evaluation; featured in Yu et al. [20] Fig. 2e and used by MAQC/SEQC consortia. *Applicability:* directly applicable; works even when batch and biology are confounded because it partitions variance without requiring balanced groups. *Implementation:* R package `pvca` (Bioconductor). *Studies in overview articles:* Yu et al. [20] demonstrate on Quartet RNA reference data; Li et al. [104 in [20]] is the foundational reference.

2. **DEG recovery against known markers (TPR, FPR, MCC).** Compare the post-correction DEG list against established trimester marker genes (e.g. CGA, CGB, LGALS13/PP13, PSG family, LEP for term; EGFR, TEAD4, CDX2 for early placenta). Compute sensitivity, false-positive rate, and Matthews correlation coefficient. *Popularity:* high; Yu et al. [20] dedicate a full subsection to DEG-based evaluation (p. 16). *Applicability:* strongly applicable; placental biology has well-established trimester markers, so a ground-truth gene set is available without needing spike-ins. *Studies:* Nygaard et al. [32 in [20]], Luo et al. [36 in [20]].

3. **Correlation before/after harmonisation.** Compute gene–gene co-expression correlations within each batch before and after correction. Good correction increases cross-batch

correlation while preserving within-batch structure. *Popularity:* high; a standard quantitative endpoint in batch-correction benchmarks [20]. *Applicability:* directly applicable even with fully confounded batches — gene–gene correlation structure should converge across studies regardless of which trimesters they contain.

4. **Cross-dataset classification (SVM, PAM, logistic regression).** Train a classifier on one multi-trimester study, predict trimester labels in another. High cross-study accuracy = batch effect successfully removed. A *dual classifier* approach (one for biology, one for batch) is most informative. *Popularity:* high; widely used in batch-correction benchmarks. Recommended by Yu et al. [20] and Adamer et al. [21]. *Applicability:* applicable for study pairs that share trimesters; not computable for single-trimester studies. *Studies:* Adamer et al. [21] use balanced accuracy and F1; Yu et al. [20] list cross-batch predictive modelling in Table 2.

5. **DRS (Distance Ratio Score).** Log-ratio of distance to closest same-biology/different-batch sample vs. closest different-biology sample. High DRS = samples cluster by biology, not batch. *Popularity:* moderate; listed in Yu et al. [20] Table 2 specifically for microarray data; also used in Adamer et al. [21]. *Applicability:* computable only for samples whose biological group appears in multiple batches; usable as a sanity check on the non-confounded subset. *Studies:* Hornung et al. [106 in [20]] (original reference); Adamer et al. [21].

6. **SNR (Signal-to-Noise Ratio, PCA-based).** Ratio of average distance between biological groups to average distance between technical replicates in PCA space. Higher SNR = better biological signal relative to technical noise. *Popularity:* moderate; proposed by Yu et al. [20] with broad data-type applicability. *Applicability:* requires technical replicates in the original formulation; can be adapted using within-trimester within-study distances as "noise" and between-trimester distances as "signal." *Studies:* Yu et al. [20] demonstrate on their Quartet RNA reference dataset.

Metrics to avoid on confounded designs.

- **kBET alone** — Yu et al. [20] explicitly warn that kBET fails when batch proportions are highly confounded with biology.
- **Batch silhouette width alone** — in a confounded design, perfect batch mixing would require destroying biological signal. Meaningful only when paired with a biology-preservation metric.
- **PCA visual inspection alone** — visualisation provides only a qualitative impression and cannot replace quantitative metrics [20].

Recommended minimal set for the placenta pipeline. We recommend a combination of (1) PVCA variance decomposition before/after correction, (2) DEG recovery against established placental trimester markers, and (3) cross-study gene–gene correlation convergence. These three metrics are well-established, straightforward to implement in R, and do not require balanced batch–biology design.

3 Placenta-specific considerations for selected methods

This section provides detailed discussion of methods most relevant to the placenta integration problem, focusing on practical caveats not captured in the overview table.

3.1 SVA: risk of biology absorption

SVA [4] estimates latent surrogate variables from residuals after removing the biological model. Its identifiability rests on the assumption that a substantial subset of genes is unaffected by the biological covariate. With placenta data where one batch contains a single trimester, the leading residual direction is essentially "this study," and any data-driven surrogate-variable estimator risks capturing trimester structure inside its surrogate factor. We recommend that on this collection SVA be used *instead of* ComBat (never on top of it) and that the recovered surrogate variables always be plotted against the trimester labels before being trusted.

3.2 RUV: practical recipe for the placenta collection

RUV [5, 6] is the most direct ComBat replacement for unbalanced placental data, provided a sensible negative-control gene list can be assembled. A reasonable starting point is housekeeping genes from the Eisenberg/Levanon list intersected with genes that have low variance across the trimester axis in a balanced reference study.

1. Pick negative-control genes (housekeeping, or empirically invariant in a balanced reference study).
2. Run `RUVg(set, controls, k)` with k chosen by the elbow of the singular values, capped at the number of batches minus one.
3. Add the `W` factors as covariates to the limma design alongside the trimester contrast.
4. Plot `W` against trimester to verify that the surrogate factors are *not* just trimester re-labelled.

3.3 Reference-batch ComBat variants

Walker et al. [7] proposed including identical pooled-RNA reference samples in every batch. Varma's BESC [8] precomputes "batch effect signatures" on a large reference cohort and subtracts them per sample. ComBat-ref [10] auto-selects the reference batch by minimum dispersion but targets RNA-seq counts (negative binomial model); for log-transformed microarray matrices the relevant analogues are Walker-style anchored ComBat (`sva::ComBat(ref.batch=)`) or BESC.

3.4 MNN / Harmony: bulk-data caveats

MNN [11] and Harmony [12] require *some* biological overlap between batches but not the full factorial. The downside is that MNN was designed for thousands of cells, not tens of bulk samples, and variance estimates driving the neighbour search are noisy on bulk data. We recommend MNN/Harmony only as a sanity check, not as the sole correction.

3.5 Cross-platform integration

Turnbull et al. [13] showed that Affymetrix and Illumina intensities can be merged at the probe level after within-platform normalisation; ComBat and XPN outperformed mean-centring and DWD. Almeida-de-Macedo et al. [14] found that pooled correlation estimates are biased when group sizes are heterogeneous, motivating per-study summaries over direct pooling.

3.6 Matrix dissection: HarmonizR

Voß et al. [16] introduced HarmonizR, which applies batch correction to incomplete matrices *without* imputation via *matrix dissection*: the combined matrix is split into sub-matrices sharing common non-missing patterns, ComBat or `removeBatchEffect()` is applied to each sub-matrix independently, and the results are reassembled. A follow-up paper [17] added blocking and singular-feature handling. Available as a Bioconductor R package (`HarmonizR`), this avoids the chicken-and-egg problem of needing batch correction before imputation but imputation before batch correction.

3.7 Batch-sensitized imputation

Goh et al. [18] compared three imputation strategies relative to batch correction: **M1** (global, all samples), **M2** (within-batch), and **M3** (cross-batch). M2 consistently preserved within-batch variance and produced lower RMSE after ComBat, while M1 and M3 increased sample variance. They also identified *batch effect associated missing values* (BEAMs) — missingness correlated with the batch factor — making global imputation particularly problematic for our collection where gene coverage differs by platform. For the placenta pipeline, this motivates imputing per-dataset (M2) before merging rather than applying softImpute to the full merged matrix.

3.8 Imputation as a complement, not a replacement

The main manuscript advocates matrix-completion imputation (*softImpute*, k-NN) followed by ComBat. This addresses the *coverage* problem (different gene catalogues) but not the *design imbalance* problem (different trimesters per study). The two are independent: imputing missing genes does not give ComBat new information about the trimester–batch confound. Any method in this supplement can be applied to the imputed matrix exactly as to the raw merged matrix.

4 Recommendations for the placenta pipeline

We suggest the following layered strategy, ordered from cheapest to most invasive, so that each layer can be added without rebuilding the previous one:

1. **Document the imbalance.** Tabulate, for every contrast of interest, which studies contribute samples from which trimester. Any contrast for which one trimester comes from disjoint studies is intrinsically risky and should be flagged in the results.

2. **Replace "ComBat → limma" with "limma with batch in the design."** This is the Nygaard/Towfic recommendation [2, 3]. It is one line of R and removes the largest source of bias for free, at the cost of some power when the design happens to be balanced.

3. **Run RUV with negative-control genes in parallel.** Use the imputed, merged expression matrix as input. Restrict to housekeeping genes intersected with empirically invariant genes from a balanced reference study. Compare the resulting DEG list against the ComBat-in-design list and report the intersection.

4. **Do not chain ComBat → RUV (or any two batch corrections) naively.** Li et al. [37] showed that sequential two-step correction inflates variance and distorts inference because the second step ignores the uncertainty introduced by the first. If both location-scale and latent-factor adjustment are desired, use a joint model such as FAbatch [38], which integrates both steps, or the ComBat+Cor framework of Li et al. that embeds the correction directly in the downstream model.

5. **If a single dataset spans the broadest range of trimesters, try reference-batch ComBat (or ComBat-ref) with that dataset as the anchor.** Use this as a third comparison point, not as the primary correction.

6. **Avoid SVA on this collection unless the residual structure has been inspected first.** The risk that SVA absorbs trimester into a surrogate variable is real and hard to detect without a balanced ground-truth subset.

7. **Consider HarmonizR as an alternative to imputation + ComBat.** HarmonizR [16] applies ComBat to sub-matrices sharing common coverage patterns, bypassing imputation entirely. This avoids the imputation–batch-correction confound documented by Goh et al. [18] and is directly applicable to the block-structured missingness in the placenta collection.

8. **If imputation is used, prefer batch-sensitized imputation (M2).** Following Goh et al. [18], impute within each dataset separately before merging, rather than imputing on the combined matrix. This preserves within-batch variance structure and improves subsequent batch correction.

The above strategy preserves the existing imputation step — which addresses a different problem (cross-platform gene coverage) — and supplements ComBat with methods that do not require a balanced design.

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